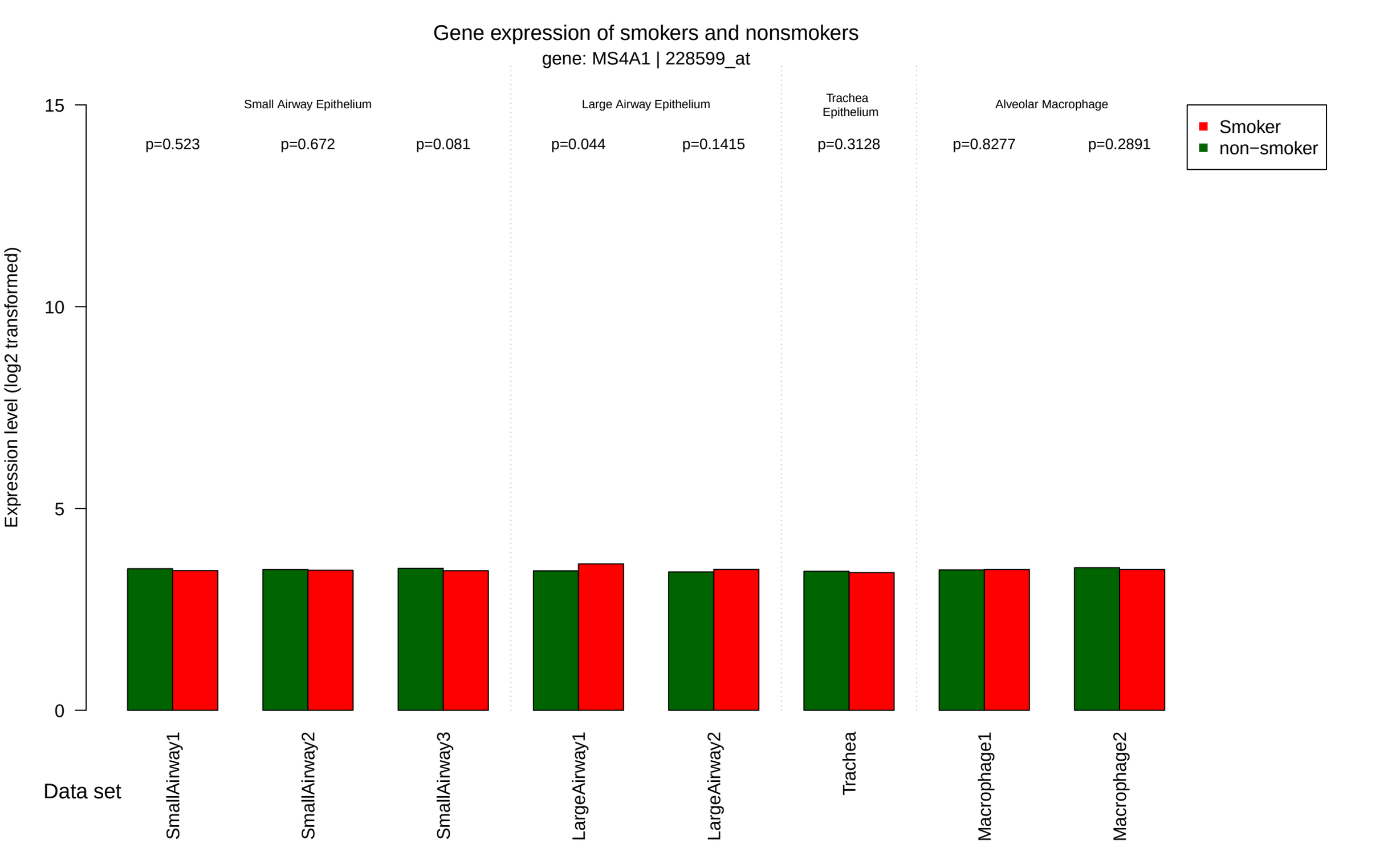
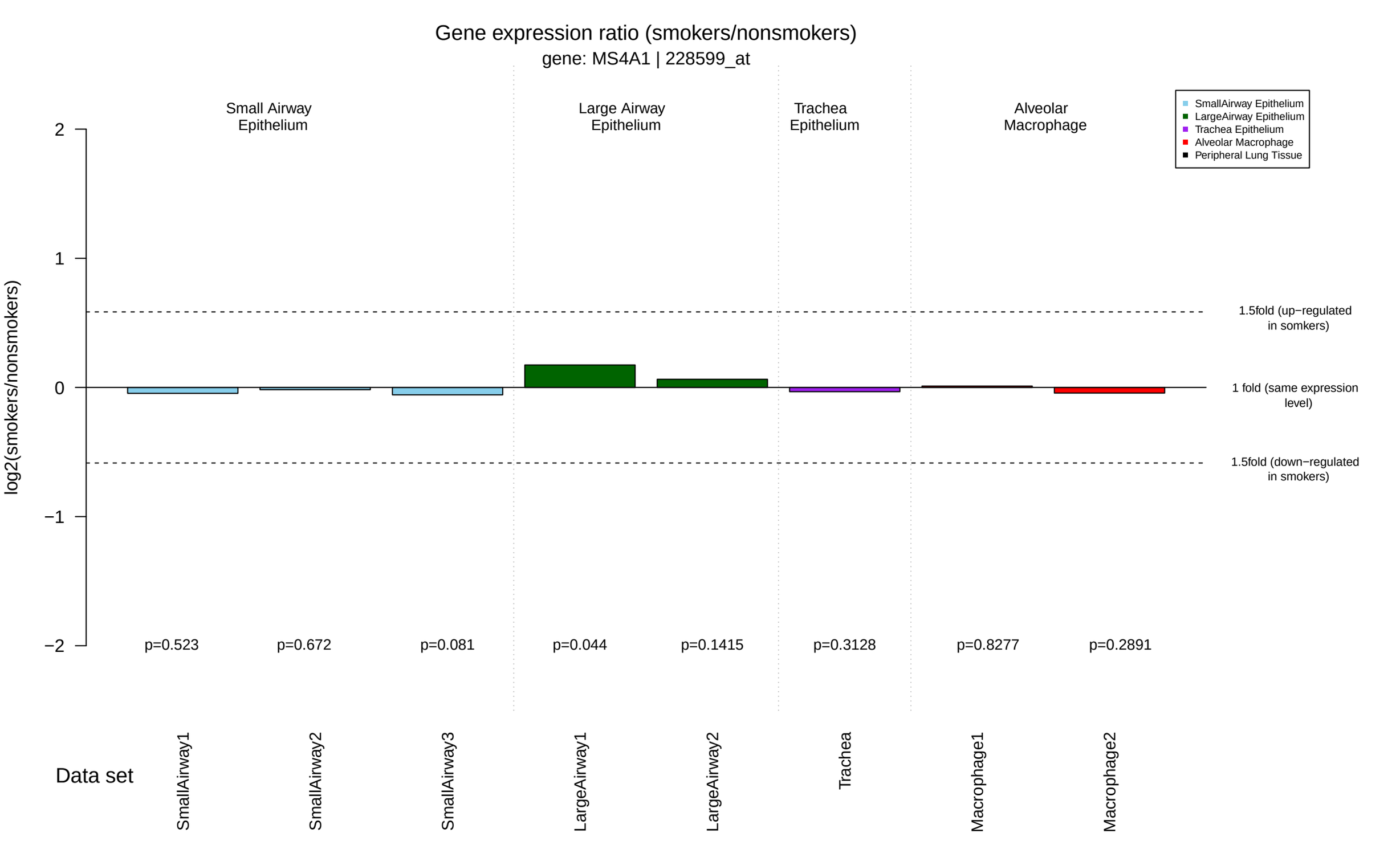
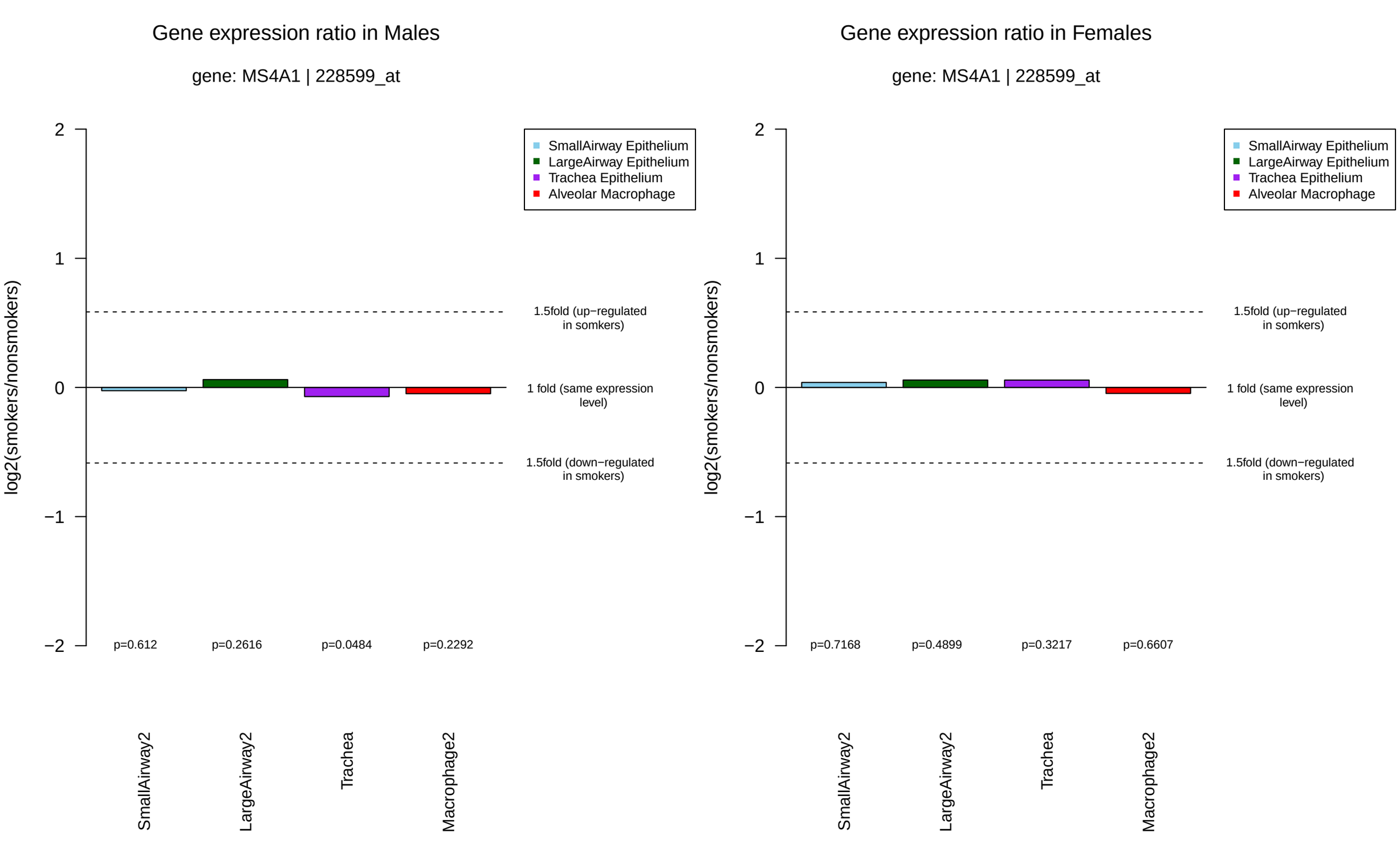
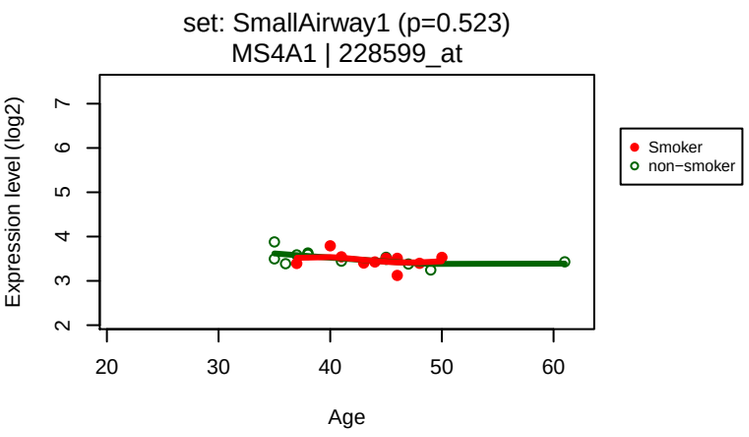
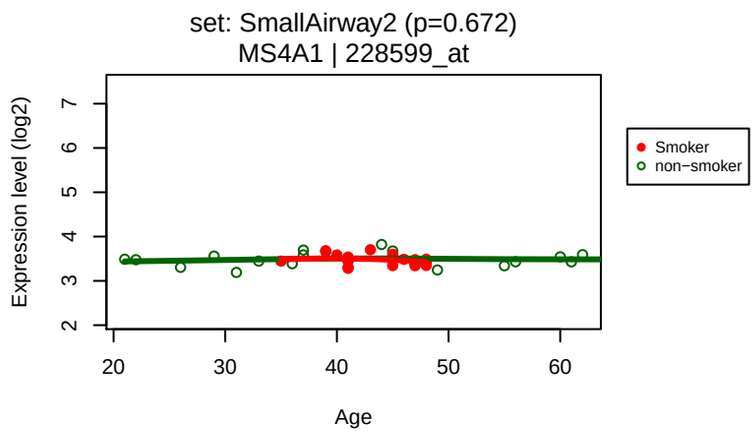
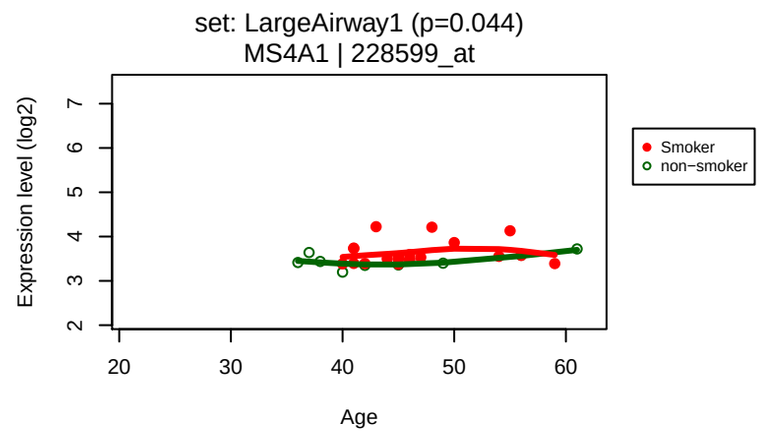

Gene expression of smokers and nonsmokers
gene: MS4A1 | 228599_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.523
p=0.672
p=0.081
p=0.044
p=0.1415
p=0.3128
p=0.8277
p=0.2891
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

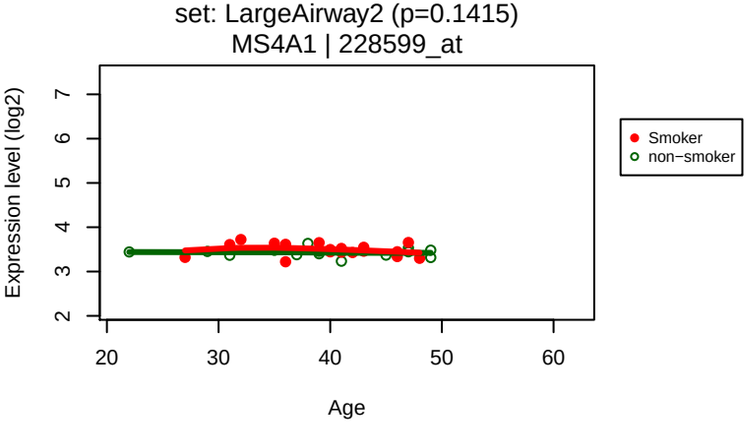

Gene expression ratio (smokers/nonsmokers)
gene: MS4A1 | 228599_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.523
p=0.672
p=0.081
p=0.044
p=0.1415
p=0.3128
p=0.8277
p=0.2891
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

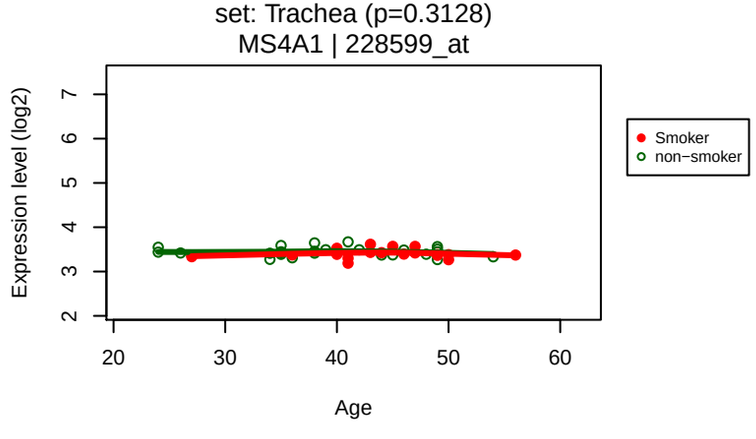

Gene expression ratio in Males
gene: MS4A1 | 228599_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.612
p=0.2616
p=0.0484
p=0.2292
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: MS4A1 | 228599_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.7168
p=0.4899
p=0.3217
p=0.6607
SmallAirway2
LargeAirway2
Trachea
Macrophage2

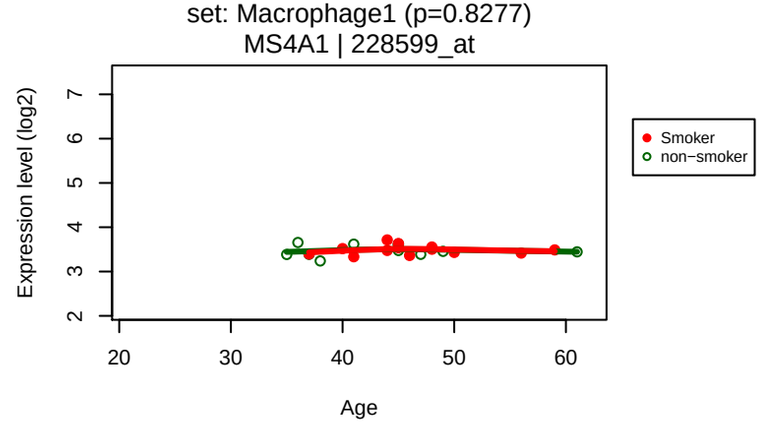

set: SmallAirway1 (p=0.523)
MS4A1 | 228599_at
Expression level (log2)
Age
Smoker
non−smoker

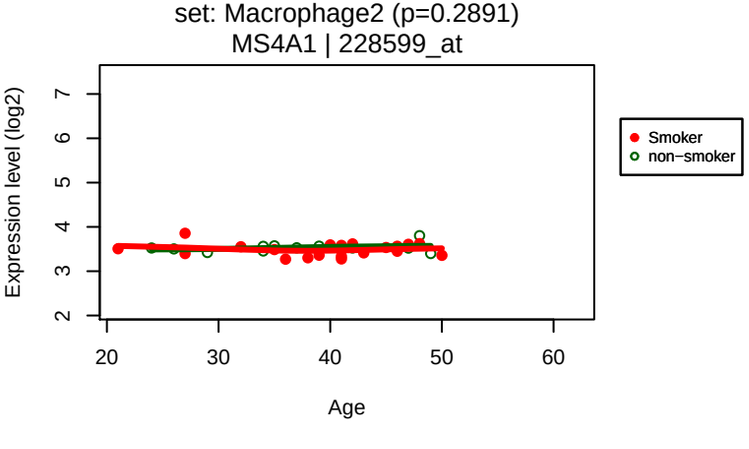

set: SmallAirway2 (p=0.672)
MS4A1 | 228599_at
Expression level (log2)
Age
Smoker
non−smoker

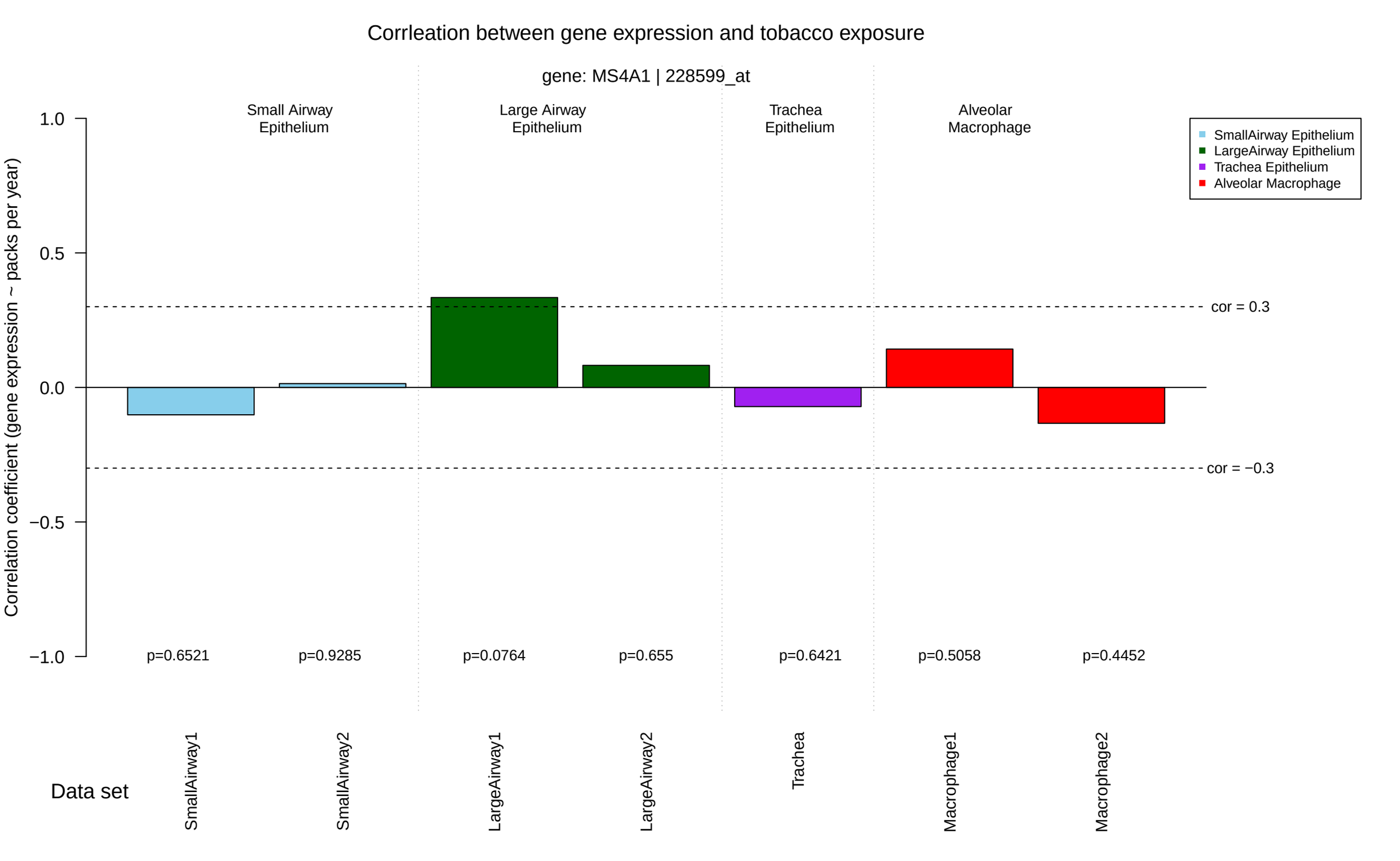

set: LargeAirway1 (p=0.044)
MS4A1 | 228599_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.1415)
MS4A1 | 228599_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.3128)
MS4A1 | 228599_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.8277)
MS4A1 | 228599_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.2891)
MS4A1 | 228599_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: MS4A1 | 228599_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.6521
p=0.9285
p=0.0764
p=0.655
p=0.6421
p=0.5058
p=0.4452
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2